PRODUCT REQUIREMENTS DOCUMENT

R3 v4 — AI-Native Digital Audio Workstation

Version 3.0 — Investor-Grade | UI-Verified | Engineering-Contracted

Confidential — Not for Distribution

DOCUMENT CONTROL

Field	Value
Version	3.0
Status	Active — MVP Build Phase
Last Updated	2026 Q1
Classification	Confidential
Audience	Engineering, Investors, Design, QA
Owner	Ty (Founder / Lead Engineer)
Stack Ref	pnpm monorepo — React/Vite + Express + tRPC + Drizzle + PostgreSQL + WebSocket + WebGPU

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0. EXECUTIVE SUMMARY

R3 v4 is an AI-native Digital Audio Workstation built for a market that existing tools have failed: DJs and music creators who need professional-quality results without professional-level technical skill or time investment.

The global music production software market is valued at $1.4B (2024) and growing at 8.2% CAGR. No dominant player in this market has shipped a genuinely AI-native workflow layer. Ableton, Logic, and Rekordbox remain fundamentally unchanged in their core interaction model for over a decade.

R3's differentiation is architectural, not cosmetic. The Low-Latency Processing Transition Engine (LLPTE) is a named, inspectable, real-time AI pipeline with confirmed sub-15ms inference latency — not a post-processing feature bolted onto an existing DAW. Every AI decision is transparent, overridable, and quantifiably faster than its manual equivalent.

The MVP is defined, scoped, and partially built. AI Auto-Leveling and Smart Transitions are implemented with 42 combined Vitest test cases across the LLPTE layer. The UI is shipped and verified. The ask is funding to close the gap between the technical foundation and the commercial product.

The one-sentence pitch: R3 makes professional-quality mixing accessible to anyone with a track and a deadline — and proves it with a number on screen.

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1. PRODUCT SUMMARY & VISION

Identity

Field	Detail
Product Name	R3 v4
Category	AI-Native Digital Audio Workstation (DAW)
Build Target	TypeScript pnpm Monorepo
Frontend	React + Vite (`client/`)
Backend	Express + tRPC + WebSocket (`server/`)
Database	PostgreSQL via Drizzle ORM
AI Pipeline	LLPTE — 6 monorepo packages (`packages/llpte-*`)
Rendering	WebGPU + Three.js / React Three Fiber
Auth	bcrypt + JWT

Payments	Stripe v20 + LemonSqueezy
Testing	Vitest — 42 cases across LLPTE layers
Deployment	Docker + Railway (`railway.toml`)
Design System	Tailwind CSS + Radix UI

Vision Statement

The future of music creation is not a smarter manual interface. It is a system that understands audio as well as the engineer sitting next to you — one that can hear what you cannot, adjust what you would have missed, and do it before you knew it was a problem.

R3 does not want to be the tool professionals use when they have time. It wants to be the tool everyone uses because it saves the time they do not have.

Core Thesis

Traditional DAWs are manual, high-skill, and time-intensive. R3 introduces LLPTE as a native AI layer that actively assists with mixing, transitions, and arrangement in real time. Every AI decision is transparent, overridable, and measurably faster than its manual equivalent.

The key insight: the value is not in AI making better decisions than humans. The value is in AI making good-enough decisions instantly, so humans can spend their time on the creative decisions only they can make.

Design Philosophy

R3's visual identity is built on five principles confirmed in the shipped UI:

1. **Dark by default.** The application lives in the studio. Every background is zinc-950 or darker.
2. **Thin borders, maximum information.** 1px borders at 6% opacity. Zero decorative chrome.
3. **Neon as signal, not decoration.** Cyan means active. Violet means AI. Amber means warning. Red means danger. These are never reversed.
4. **JetBrains Mono for data, Inter for language.** BPM, latency, dB values, frequencies — all monospace. All labels, navigation, suggestions — Inter.
5. **Every AI action leaves a visible trace.** Ghost knobs, violet overlays, confidence

scores, toast notifications. Nothing the AI does is invisible.

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2. MARKET OPPORTUNITY

Total Addressable Market (TAM)

Segment	Size	Source
Global Music Production Software	$1.4B (2024)	Grand View Research
CAGR	8.2% through 2030	Grand View Research
Projected Market (2030)	$2.3B	Calculated
DJ Software Segment	$380M	Statista
Content Creator Audio Tools	$620M	IBISWorld

Serviceable Addressable Market (SAM)

R3's initial wedge targets two creator segments that are underserved by existing tools:

- **Digital DJs:** 4.2M active DJs globally (DJPA 2023). 1.8M use software-based DJ tools. Average software spend: $180/year. SAM from DJs alone: **$324M**.
- **Content Creators requiring audio:** 50M+ active content creators (YouTube, TikTok, Podcasts). Approximately 12M spend money on audio tools. Average audio software spend: $85/year. SAM from creators: **$1.02B**.

Combined SAM: approximately **$1.34B**.

Serviceable Obtainable Market (SOM)

Year 1 target: 5,000 paying users at $20/month = **$1.2M ARR**. Year 2 target: 25,000 paying users at $20/month + enterprise tier = **$7.5M ARR**. Year 3 target: 100,000 paying users + B2B API licensing = **$35M ARR**.

Why Now

Three forces converge in 2025–2026 to make R3 viable and necessary:

1. **WebGPU is production-ready.** Real-time GPU-accelerated audio visualization and AI

inference is now possible in a browser-native desktop application without a plugin. This removes the barrier that kept AI out of web-based audio tools.

2. **Quantized inference is fast enough.** Sub-15ms inference latency on consumer hardware was not achievable 24 months ago without specialized hardware. It is achievable today. R3 ships at 10ms.
3. **The creator economy demands professional output.** The bar for audio quality in short-form content has risen dramatically. Audiences trained by Spotify and Apple Music reject amateur mixes. Creators know this — they just do not know how to fix it. R3 does.

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3. COMPETITIVE LANDSCAPE

Direct Competitors

Ableton Live 12

Dimension	Ableton Live 12	R3 v4
AI Integration	None (Max for Live patches only)	Native LLPTE pipeline
Learning Curve	6–12 months to proficiency	Target: 30 minutes to first AI-assisted mix
Price	\$99–\$749 one-time	\$20/month (SaaS)
Target User	Professional producers	DJs + creators (entry to mid)
Latency	ASIO/CoreAudio dependent	≤10ms guaranteed
AI Transparency	N/A	Ghost knobs, confidence scores, node graph
Time Savings Tracking	None	Native — session-level and action-level

Native Instruments Traktor Pro 4

Dimension	Traktor Pro 4	R3 v4

AI Mixing	None	Full LLPTE pipeline
Transition Automation	Manual cue points only	AI-generated transition curves
Frequency Conflict Detection	None	Spectral Analyzer → Mix Suggestion System
Business Model	$99 one-time	SaaS — monthly with tier escalation
Platform	Desktop only	Desktop-first; web-ready architecture

Rekordbox 7

Dimension	Rekordbox 7	R3 v4
AI Features	"Intelligent" track analysis (BPM/key only)	Real-time mix decisions, transitions, EQ
Hardware Lock-in	Pioneer ecosystem required for full features	Hardware agnostic
Subscription	$15/month (Performance plan)	$20/month with full AI
Open Ecosystem	No	tRPC API — extensible

Emerging AI Competitors

Product	Status	Gap vs R3
Soundraw	Generative, not mixing	Does not address mix quality
Aiva	Composition only	Entirely different use case
Landr	Mastering only (post-production)	No real-time mixing assistance
Adobe Podcast	Voice only	No music production capability
Udio / Suno	Generation only	No DJ/mixing workflow

Competitive Moat

R3's moat is not the AI model. The moat is the **LLPTE architecture**: a named, versioned, inspectable real-time pipeline that is embedded in the product's visual identity.

Competitors cannot copy the node graph, the ghost knobs, or the Time Savings panel without rebuilding their entire product architecture. The switching cost grows with every session's worth of saved preferences and AI calibration data.

Secondary moat: **data flywheel.** Every accepted AI suggestion is a labeled training signal. At 5,000 users with 3 sessions/week, R3 accumulates approximately 780,000 labeled audio engineering decisions per month. No competitor has this dataset.

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4. BUSINESS MODEL & MONETIZATION

Subscription Tiers

Starter — Free

Designed to convert creators who have never paid for audio software.

- 4 simultaneous tracks
- AI Auto-Leveling (limited: 5 uses per session)
- Basic transition: crossfade only
- Time Savings Tracking: session summary only (no export)
- Watermark on exported audio
- Community support only

Pro — $20/month

The primary commercial tier. DJs and serious creators.

- Unlimited tracks
- Full AI Auto-Leveling (unlimited)
- All 7 Smart Transition types
- Full Mix Suggestion System
- LLPTE node graph (visible and inspectable)
- Time Savings Tracking with export (PNG + JSON)
- Effects Rack: all 6 modules
- Priority WebSocket inference queue

- Email support
- Pro badge visible in UI (social proof within product)

Studio — $60/month (Post-MVP)

Professional producers and small studios.

- Everything in Pro
- Multi-user session collaboration (real-time WebSocket)
- Custom AI model fine-tuning on user's own mix history
- API access (tRPC endpoints exposed)
- White-label export (no R3 branding)
- Dedicated support channel

Enterprise / API — Custom Pricing (Post-MVP)

Platforms, labels, and hardware manufacturers.

- LLPTE pipeline exposed as a REST/WebSocket API
- Custom inference model training on proprietary catalog
- SLA-backed latency guarantees
- On-premise deployment option (Docker)

Revenue Projections (Conservative)

Quarter	Paying Users	MRR	ARR
Q1 Post-Launch	500	$10,000	$120,000
Q2	1,500	$30,000	$360,000
Q3	3,500	$70,000	$840,000
Q4	6,000	$120,000	$1,440,000
Year 2 Q4	25,000	$500,000	$6,000,000

Unit Economics

Metric	Target
Customer Acquisition Cost (CAC)	≤$35 (content-led, not paid)
Lifetime Value (LTV) at $20/month	$240 (12-month average retention)
LTV:CAC Ratio	≥7:1
Gross Margin	82% (SaaS infrastructure + AI inference cost)
Monthly Churn Target	≤4%
Payback Period	≤2 months

Monetization Hooks Inside the Product

Every free-tier limitation is a visible, non-punishing upgrade prompt:

- Track limit reached → "Add unlimited tracks with R3 Pro" — shown as a dimmed track slot with a Pro badge
- Export watermark → "Remove watermark with R3 Pro" — shown in export dialog
- AI uses exhausted → "Unlock unlimited AI leveling" — shown in the toast notification system
- Session export disabled → "Export your time savings stats" — shown in session summary panel

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5. TARGET USERS — DEEP PERSONAS

Persona 1: Marcus — The Working DJ

Demographics: 28 years old, Chicago. DJs 3–4 nights per week at mid-tier venues ($200–$500/night). Has been DJing for 6 years. Uses Rekordbox + Pioneer CDJ setup live, Ableton for prep work at home.

Technical Skill: High for performance, medium for production. Can beatmatch manually but prefers sync. Knows what a good transition sounds like but struggles to execute complex multi-effect transitions consistently.

Jobs to Be Done:

- Prepare a 2-hour set in under 4 hours, not 8

- Guarantee transition quality even when playing unfamiliar tracks
- Detect energy arc problems before they happen on the floor
- Impress venues enough to be re-booked

Current Pain Points:

- Ableton prep is slow — manually setting cue points, testing transitions, adjusting EQ per track takes the majority of his prep time
- No tool tells him when two tracks are going to clash spectrally before he plays them
- His mixes sound inconsistent — good nights and bad nights depend on energy and time, not skill

How R3 Solves Marcus's Problems:

- LLPTE detects phrase boundaries automatically — Marcus sets the transition type, not the timing
- Spectral Analyzer flags frequency clashes before playback — he sees the conflict on the timeline, not hears it on the floor
- AI Auto-Leveling balances track volumes in the first 2 bars — his mixes start clean every time
- Time Savings panel confirms prep time reduction — Marcus can show his manager he prepared a full set in 3 hours

Acquisition Channel: DJ forums (r/DJs, DJTT), short-form video showing "Set prep in 1 hour instead of 4."

Willingness to Pay: $20/month without hesitation if the transition quality is provably better.

Persona 2: Priya — The Content Creator

Demographics: 24 years old, Austin. Full-time YouTube creator, 180K subscribers. Produces 2 videos per week covering lifestyle and travel. Uses Adobe Premiere for video, no dedicated audio tool.

Technical Skill: Near zero for audio. Can cut and arrange video perfectly. Treats audio as an afterthought until a video gets a comment saying "the music is too loud."

Jobs to Be Done:

- Make her background music sound like it was mixed by a professional
- Never get a comment about audio quality again
- Spend less than 20 minutes on audio per video
- Have music that ducking-compensates automatically when she speaks

Current Pain Points:

- Her current workflow: drag music into Premiere, drag a volume keyframe down when she speaks, export. That's it. It sounds amateur.
- She has tried GarageBand once. She did not understand what any of the knobs did. She closed it.
- She cannot hear the difference between 128kbps and 320kbps audio but her audience can

How R3 Solves Priya's Problems:

- AI Auto-Leveling handles all gain staging — she does not need to know what LUFS means
- Mix Suggestion System catches frequency masking between her voice and the background music automatically
- The interface shows confidence scores, not technical parameters — `94% confident this sounds better` is something she can act on
- Time Savings panel gives her a number for her end-of-month productivity report

Acquisition Channel: Creator economy newsletters, "How I make my YouTube audio sound professional in 10 minutes" video.

Willingness to Pay: $20/month if it saves her 1 hour per week. She values her time at $100+/hour.

Persona 3: Deon — The Bedroom Producer

Demographics: 19 years old, Atlanta. Makes beats for SoundCloud and wants to start placing them with artists. Has FL Studio. Has watched 400+ hours of YouTube production tutorials.

Technical Skill: High for sound design, low for mixing and mastering. Can build a beat from scratch in 2 hours. Cannot make it sound like it was mixed by a professional.

Jobs to Be Done:

- Close the gap between his beats and the professionally mixed tracks he hears on streaming platforms
- Get feedback on specific problems in his mix, not generic advice
- Learn what he is doing wrong through the AI's decisions, not tutorials
- Have something to show an A&R contact that sounds finished

Current Pain Points:

- His mixes are "muddy" — he knows it, cannot fix it
- Frequency masking between bass and kick is the core problem — he has read about it, cannot hear it
- He spends 4 hours mixing a beat and it still sounds worse than a 2-hour beat by a professional

How R3 Solves Deon's Problems:

- Spectral Analyzer visualizes the frequency clash between KICK and 808 BASS in real time — he can see what he could not hear
- Mix Suggestion System says `"Cut low-end on SYNTH LEAD below 80Hz — clashing with 808 BASS"` — specific, actionable, educational
- The AI teaches him as it assists him — each accepted suggestion is a lesson
- LLPTE Confidence score shows him how certain the AI is — he develops audio intuition by proxy

Acquisition Channel: SoundCloud community, beat-making subreddits, "R3 made my beat sound like a hit" TikTok.

Willingness to Pay: $20/month if it gets him a placement or a serious compliment on mix quality.

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6. OBJECTIVES & SUCCESS CRITERIA

Strategic Objectives (Ranked)

1. Demonstrate that AI can produce a measurably better mix than a human in less time — proven by the Time Savings panel and before/after audio comparison.

2. Establish LLPTE as a named, defensible, visible AI architecture that differentiates R3 from every competitor.
3. Achieve 65+ NPS among alpha users within 60 days of private launch.
4. Reach $120K ARR within 12 months of public launch.
5. Collect 1M+ labeled AI decision signals for model improvement within 18 months.

Feature-Level Success Criteria

AI Auto-Leveling

Metric	Target	Measurement Method
Clipping event reduction	≥70%	VU meter clip event log, per session
AI adjustment acceptance rate	≥65%	Accept/reject event in analytics schema
Inference latency	≤15ms sustained	LLPTE Core badge — live, sampled every 100ms
Audio artifact rate	0 perceptible artifacts	QA listening test at 48kHz/24bit
First AI action time	Within 2 bars of playback	Automated test against mock audio input

Smart Transitions

Metric	Target	Measurement Method
Transition setup time reduction	≥50%	Manual timing vs AI timing per session log
Suggested transition acceptance	≥70% without edits	Edit event log post-AI generation
Transition execution latency	≤20ms	Transition Graph node tick time
Phrase boundary detection accuracy	≥90%	Labeled test set of 200 boundary positions
Precompute lead time	≥2 bars before boundary	Timing log vs playhead position

Mix Suggestion System

Metric	Target	Measurement Method
Suggestion acceptance rate	≥40%	Accept/reject analytics
False positive rate	≤10%	User override without modification
Suggestion latency	≤500ms from trigger	Event timestamp diff
Suggestions per session	≥3 visible	Session analytics
Manual EQ reduction	≥30%	Mixer interaction event log

LLPTE Core

Metric	Target	Current (Verified)
Inference latency	≤15ms	10ms (live badge)
Node tick time	≤1ms	0.8ms (tooltip)
Active edges (stable)	≥800	847 (confirmed)
Track ceiling	≥8	8 (MVP)
Pipeline uptime	100% during demo	QA checklist

Time Savings Tracking

Metric	Target	Measurement Method
Panel adoption	≥80% of sessions open it	UI interaction analytics
Demo use rate	100% of investor demos	Pre-demo checklist
Export function reliability	100% success	Automated test
Displayed savings believability	≥85% find it credible	Alpha user survey

Overall Product

Metric	Target

NPS (alpha users, day 30)	≥65
Day 7 retention	≥60%
Day 30 retention	≥40%
Sessions per week per active user	≥3
Average session length	≥25 minutes
Feature discovery rate (AI features)	≥60% of new users use ≥1 AI feature in first session

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7. UI ARCHITECTURE — FIVE ZONES (VERIFIED FROM SHIPPED UI)

All specifications in this section are derived from the verified shipped interface screenshot and cross-referenced against the monorepo source tree.

Zone 1 — Top Navigation Bar (48px, full-width)

Background: Absolute black. Zero transparency.

Components (left to right)

R3 Wordmark

- Font: JetBrains Mono, weight 700, letter-spacing 0.2em
- Color: Pure white
- Version badge: `v4.2.1` — zinc-700 pill, JetBrains Mono 11px

Transport Controls (center)

- Icon strip: skip-back ⏮️ , stop ⏹️ , record ⏺️ , play ▶ , skip-forward ⏭️
- Play button state: cyan glow `#00F5FF` when active — this is the visual confirmation that audio engine is running
- Icon size: 16×16px SVG
- Touch target: 32×32px minimum

BPM Counter

- Font: JetBrains Mono
- Value: `128.00` — 2 decimal places always shown
- Tap-tempo button adjacent — resets BPM to tapped interval
- Engineering requirement: BPM counter must update via AudioParam scheduling, not UI frame rate — decoupled from React render cycle

Grid Controls

- Time signature: `4/4` — clickable, opens dropdown (3/4, 4/4, 5/4, 6/8, 7/8)
- Grid snap: `1/16` — dropdown (1/1 through 1/32, triplets)
- Sync indicator: small dot, tempo-locked flash at BPM rate

Right Side

- CPU/memory meter: 80px horizontal bar, green fill — updates every 500ms
- User avatar: 32px circle, user profile photo from `server/storage.ts`
- Pro badge: `Pro` violet pill — data from `shared/schema-subscription.ts`
- Settings gear: links to global settings panel

Engineering Notes

- Transport state broadcast via WebSocket to all connected clients — `server/ws/`
- BPM state stored in Redux slice — `@reduxjs/toolkit` package confirmed in node_modules
- Pro badge driven by subscription tier from database — Stripe webhook updates tier field

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Zone 2 — Left Sidebar — Project & Sample Browser (240px)

Background: zinc-900. Right border: 1px zinc-800.

Components (top to bottom)

Search Field

- Placeholder: `search samples, midi, presets...`
- Font: JetBrains Mono 13px, zinc-400

- Filter icon: right-aligned inside input
- Behavior: debounced 200ms — queries tRPC `sampleSearch` procedure

File Tree

- Root nodes: PROJECTS, SAMPLES, MIDI, PRESETS, UPLOADS
- Chevron collapse/expand per node — persisted in localStorage
- Expanded state under SAMPLES: `kicks/` , `808s/` , `atmospheres/` , `vocal/`
- File-type icons: waveform SVG for audio, MIDI plug SVG for MIDI files, folder icon for directories
- Node height: 28px. Font: Inter 13px, zinc-400

Active File Highlight

- Example active file: `Orbital_Kick_01.wav`
- Style: cyan left-border 2px `#00F5FF` , text zinc-100, background zinc-800

Mini Waveform Preview

- Height: 48px
- Waveform color: electric blue `#3B82F6`
- Micro play button: 16px, right-aligned
- Behavior: renders on hover over any audio file — debounced 100ms

File Metadata

- Format: `44.1kHz / 24bit / 2.3s`
- Font: JetBrains Mono 11px, zinc-400

Engineering Notes

- File tree populated by tRPC procedure reading `uploads/` directory structure
- Waveform preview: server-side peak extraction, returned as Float32Array, rendered via Canvas API
- Path traversal protection: all file access validated against allowed directories — audit-confirmed
- Sample upload: `multer` v2.1.1 — confirmed in node_modules

Zone 3 — Center Arrangement / Timeline (~60% width)

Background: zinc-950. This is the dominant visual and functional zone.

Timeline Ruler

- Bar numbers in JetBrains Mono 11px, zinc-500
- Current range shown in screenshot: bars 1–30+
- Major gridlines every 4 bars: 1px zinc-800
- Minor gridlines every beat: 1px zinc-900
- Playhead: 1px solid `#00F5FF`, triangle handle at top (8px), visible across all track rows
- Playhead luminance bleed: 4px radial feather in cyan at 8% opacity — gives impression of screen phosphor

Eight Track Definitions

Track Row Structure (72px tall each)

- Left strip: 180px wide
  - Color square: 12×12px, track color
  - Track name: Inter 12px medium, zinc-100
  - Mute button: `M` — zinc-700 background, active state zinc-400
  - Solo button: `S` — same
  - Record arm button: ● — red when armed
  - VU meter: 4px wide vertical strip, green/amber/red

Track	Color	Waveform Type	Special Behaviors
KICK	`#EF4444`	Sharp transient spikes, near-silent valleys	Beat-grid aligned, transient markers
808 BASS	`#F59E0B`	Wide smooth sine-like shapes, slow decay	Pitch automation lane, portamento curve
SYNTH LEAD	`#8B5CF6`	Dense harmonic oscillations, amplitude variation	Crossfade blend zone at clip boundaries
CHORD		Sustained blocks, mixed audio	MIDI piano-roll clip type

PAD	`#06B6D4`	+ MIDI regions	supported
VOCAL CHOP	`#F43F5E`	Fragmented stutter pattern, many short clips	Right-click context menu — confirmed active
FX SEND	`#10B981`	Sparse signal, automation lane expanded	Reverb Send automation curve visible below
AI MIX	`#8B5CF6` overlay	Translucent prediction zone + dashed curve	LLPTE output lane — `94%` confidence badge
MASTER	`#64748B`	Full stereo output, 90px height	Lissajous vectorscope right side

Clip Interaction States

State	Visual Treatment
Default	Track-color waveform on zinc-950
Selected	Cyan border 1px, slightly brighter fill
AI-generated region	Translucent violet gradient, dashed boundary
Crossfade zone	Overlapping translucent blend at clip edges
Right-click active	Context menu overlay, backdrop dimmed 20%

Right-Click Context Menu (confirmed in shipped UI — VOCAL CHOP)

Option	Action	AI-Specific
Rename	Inline text edit	No
Duplicate	Clone clip to same track	No
Delete	Remove clip, shift remaining	No
Bounce to Audio	Render MIDI/automation to audio	No
Send to AI	Route clip into LLPTE pipeline	**Yes — R3 differentiator**

"Send to AI" Flow:

1. LLPTE receives clip audio buffer

2. Spectral analysis runs immediately
3. Transition type selector appears above clip
4. Violet AI region overlay renders on timeline
5. Precomputed transition curve drawn as dashed line
6. Preview button appears — auditions transition without committing
7. Accept renders the transition permanently

Timeline Minimap (right edge, 60px wide)

- Compressed overview: all 8 tracks as colored horizontal bars at 1/30 scale
- Viewport indicator: white semi-transparent rectangle showing current view window
- Click/drag minimap to scrub quickly through arrangement
- Updates every 250ms — does not block audio thread

Zone 4A — DJ Mixer / Channel Strips (bottom-left, 220px tall)

Six vertical channel strips (KICK, BASS, SYNTH, CHORD, CHORD, VOCAL, FX)

Per-Strip Anatomy (top to bottom)

Channel Header Chip

- Track label text, Inter 11px medium
- Background: track color at 20% opacity, zinc-900 base

3-Band EQ Section

- Three rotary knobs: HI, MID, LO
- Knob design: dark aluminum housing, white indicator line — photorealistic
- Each knob independently positioned
- Range: ±15dB per band
- Default: all at 12 o’clock (unity)

Gain Trim Knob

- Pre-fader level control
- Range: −∞ to +6dB
- Shows ghost position when AI Auto-Leveling is active

VU Meter Bar

- Vertical orientation
- Gradient: green (−40 to −12dBFS) → amber (−12 to −6dBFS) → red (−6 to 0dBFS)
- Peak hold indicator: 1px line held for 2 seconds before decay
- Update rate: 60fps — must not drop below 30fps under load

Channel Fader

- Vertical slider, full throw = 100mm minimum touch target
- Gray cap, zinc-800 track
- Range: −∞ to +6dB
- Double-click to reset to unity

DJ Crossfader (bottom, full width)

- Slim horizontal track, white puck
- Currently centered in screenshot
- Range: hard left to hard right, with configurable curve (linear/exponential)
- AI-assisted position: ghost indicator shows AI-suggested crossfader position during transitions

Performance Pads (bottom corners, 4×2 grid per deck)

Color	Function	Behavior
Red	Hot Cues	Jump to stored cue point
Green	Loop Points	Toggle loop on/off at marker
Blue	Sampler Triggers	Fire one-shot sample
Orange	Effect Toggles	Enable/disable effect per channel

Zone 4B — LLPTE Node Graph (bottom-right)

This is the technical proof-of-concept panel. It must be animated and live during every investor demo.

Canvas: zinc-950 background, 8px dot grid pattern in zinc-900 at 15% opacity.

Node Specifications

Node	Package	Color	Size	Special Display
Input Router	`@llpte/llpte-adapters`	Gray	Standard	Entry point label
Spectral Analyzer	`@llpte/llpte-signal`	Cyan	Standard	Mini FFT sparkline
LLPTE Core	`@llpte/llpte-ai`	Violet	Large (2×)	`inference 10ms` badge, arc progress, mini waveform
Transition Graph	`@llpte/llpte-transition-graph`	Blue	Standard	Active edge count, tick time tooltip
Output Bus	`@llpte/llpte-execution`	Emerald	Standard	Level indicator

Connector Lines

- Width: 1px
- Style: animated dashes — 5px dash, 5px gap, moving in signal direction at 30px/second
- Color: matches target node accent color
- These must animate continuously — static connectors are a demo failure condition

Tooltip Specification (hover over Transition Graph)

```
llpte-transition-graph v1.2
847 active edges
last tick 0.8ms
```

- Background: zinc-800, border zinc-700

- Font: JetBrains Mono 11px
- Delay: 200ms hover before show, no delay on hide

Signal Oscilloscope (below graph)

- Height: 40px
- Last 2 seconds of processed signal
- Two overlapping waveforms: violet (pre-AI) and cyan (post-AI)
- Update rate: 30fps
- Zero-line: zinc-700 at 1px

LLPTE Core Node Detail

- Size: 2× standard node dimensions
- Animated 3-segment arc (loading spinner variant) showing active inference cycle
- Mini waveform inside node body — last 500ms of processed signal
- `inference 10ms` badge: bottom-right of node, emerald background, JetBrains Mono 11px
- Pulsing ring animation when inference is running

Zone 5 — Right Sidebar — Effects Rack (280px)

Background: zinc-900. Left border: 1px zinc-800.

Module Card Anatomy

Each effect module is a card with:

- **Header row:** Module name (Inter 13px medium, zinc-100) + power toggle (green when on) + drag handle (⋮⋮) + expand arrow (`>`)
- **Active indicator:** 6px dot — emerald `#10B981` when on, zinc-600 when bypassed
- **Parameter section:** Rotary knobs or slider controls
- **Visualization strip:** Module-specific display (curve, waveform, meter)
- **Card background:** zinc-850 (between zinc-800 and zinc-900)

- **Card border:** 1px zinc-800, radius 8px

Module Specifications

Reverb

- Parameters: Room Size (%), Decay (seconds), Wet/Dry (%)
- Shipped values: `Size: mid-high, Dec: 3.2s, Wet: 28%`
- Visualization: frequency response decay tail — exponential curve in translucent blue on zinc-950 canvas
- Status: Active (green dot)
- Knob labels: Sat (saturation/size), Sec (decay in seconds), Fne (fine wet/dry)

Delay

- Parameters: Swel (swell/time), Amp (amplitude/feedback), Sep (separation/spread)
- Visualization: animated ping-pong dots between L and R channels at tempo-locked rate
- Status: Active (green dot)

Compressor

- Parameters: Trs (threshold), Atk (attack), S.too (sustain/release)
- Visualization: gain reduction meter — `-3.2dB GR` in amber when active
- Status: Active (green dot)

Transition Graph (LLPTE Module Card)

- Visual module mirroring Zone 4B data at compact scale
- Level indicators showing signal flow state
- Status: Active (green dot)

Output Bus (LLPTE Module Card)

- Signal routing card
- Level indicators: Left/Right channel meters
- Status: Active (green dot)

EQ

- Visualization: 80×40px frequency curve graph on zinc-950 canvas, rendered in cyan
- Default shape: Boost at 80Hz, dip at 3kHz, dip at 8kHz
- Parameter: Sep (frequency separation control)
- Status: Active (green dot)

---

8. CORE FEATURES — FULL SPECIFICATION

8.1 AI Auto-Leveling

Problem Statement

Gain staging is the most common source of amateur-sounding mixes. A track recorded at −6dBFS and a track recorded at −18dBFS will produce a mix with one element dominating before a single creative decision is made. Professional engineers spend significant time addressing this before mixing begins. R3 addresses it automatically within the first 2 bars of playback.

Technical Architecture

```
Track Audio Buffer
      │
      ▼
spectralAnalyzer (FFT, RMS, LUFS, true peak measurement)
      │
      ▼
aiMixEngine (gain decision — confidence scored)
      │
      ├── If confidence ≥ 0.65: Auto-apply via AudioParam.setTargetAtTime()
      │
      └── If confidence < 0.65: Surface as ghost knob suggestion only
```

Input Specification

Input	Type	Source	Notes
Track gain (dBFS)	Float32	AudioParam	Per track
FFT output	Float32Array	AnalyserNode	2048 bins, Hann window

RMS	Float	Computed	50ms window
LUFS	Float	ITU-R BS.1770-4	Integrated and momentary
True peak dBTP	Float	Inter-sample peak	Prevents clipping post-export
Track count	Integer	Arrangement state	1–8 for MVP

Output Specification

Output	Type	Delivery	Notes
Gain adjustment (dB)	Float	AudioParam.setTargetAtTime()	Click-free
EQ suggestion	Object	UI state → ghost knob	Not auto-applied
Confidence score	Float [0,1]	UI badge + toast	Displayed as percentage
Adjustment rationale	String	Suggestion panel	Human-readable

UI States

State	Visual Treatment
Inactive	No ghost knobs, toggle off
Analyzing (bars 1–2)	Subtle pulse on AI MIX track header
Suggestion ready	Ghost knob appears, toast fires
Accepted	Knob animates to new position, ghost dissolves
Rejected	Ghost dissolves, rejection logged
Override	Manual adjustment captured, preference delta stored

Confidence Gating Logic

```
// Confidence gate — LLPTE Core
const CONFIDENCE_THRESHOLD = 0.65;
const SUGGESTION_THRESHOLD = 0.40;
```

```
if (confidence >= CONFIDENCE_THRESHOLD) {
  applyViaAudioParam(gainAdjustment);
  logAction('auto_applied', confidence);
} else if (confidence >= SUGGESTION_THRESHOLD) {
  surfaceGhostKnob(gainAdjustment, confidence);
  logAction('suggested', confidence);
} else {
  // Below suggestion threshold — discard silently
  logAction('discarded', confidence);
}
```

Testing Requirements

- 20 Vitest test cases (implemented — confirmed in coverage report)
- Test scenarios: silence input, clipping input, single track, 8 tracks, frequency-dense input, impulse-only input
- Latency test: confirm ≤15ms inference across 1000 iterations
- Accuracy test: compare output to reference mix engineer decisions on 50 labeled tracks

---

8.2 Smart Transitions

Problem Statement

A DJ transition is a sequence of audio engineering decisions compressed into a 16–32 bar window: tempo alignment, key matching, frequency separation, energy arc management, and effect timing. Executed manually, this takes preparation time and real-time attention. Executed via LLPTE, the curve is precomputed before the boundary arrives.

Transition Type Specification

Type	Mechanism	LLPTE Node	Best For
Crossfade (linear)	Equal-power gain crossfade	transitionGraph	Same-key tracks
Crossfade (S-curve)	Sigmoidal gain curve	transitionGraph	Energy preservation
Filter Sweep	LPF close → HPF open on incoming	aiMixEngine	High-energy drops

Reverb Tail	Outgoing track reverb extended, incoming fades in underneath	aiMixEngine	Atmospheric sections
Beat Drop Alignment	Transition delayed to nearest downbeat	transitionGraph	Drop-centric sets
Echo Freeze	Outgoing track pitch-locked, incoming crossfades in	aiMixEngine	Dramatic transitions
Key-Match Crossfade	Harmonic mixing via Camelot wheel scoring	aiMixEngine	Maximum harmonic quality

Camelot Wheel Harmonic Scoring (confirmed — Smart Transitions implementation)

```
Perfect match (same key): score = 1.0
Adjacent key (±1 position): score = 0.8
Relative major/minor: score = 0.7
Compatible key (±2 positions): score = 0.5
Incompatible key: score = 0.2 (transition still offered, lower confidence)
```

LLPTE Transition Execution Timeline

```
T-2 bars: Phrase boundary detected by spectralAnalyzer
           → transitionGraph begins precomputation
T-1 bar:  Transition curve fully precomputed
           → Violet region overlay appears on timeline
           → Transition type selector appears
T-0:      User accepts or system auto-executes at confirmed boundary
           → transitionGraph executes precomputed curve
           → Execution latency: <20ms from trigger to first sample
```

UI Integration Detail

Timeline Overlay:

- Violet gradient region: `rgba(139, 92, 246, 0.15)` fill
- Dashed boundary lines: 1px violet, 4px dash / 4px gap
- Transition type label: small violet chip above region
- Duration indicator: bar count shown in JetBrains Mono

Transition Type Selector (popover):

- Appears above the transition region on timeline
- 7 type options as icon+label chips
- Selected type highlighted in violet
- Confidence score shown per type

Preview Button:

- Appears in popover
- Auditions transition in isolation — does not affect playback position
- Preview plays 4 bars before + transition + 4 bars after

8.3 Mix Suggestion System

Trigger Detection Logic

Frequency Masking Detection:

```
For each pair of tracks A, B:
  Compute FFT overlap in shared frequency bands
  If overlap energy > -12dBFS in any 1/3-octave band
    AND both tracks are active (not muted/soloed)
    AND overlap duration > 1 bar:
  → Trigger suggestion: "Cut [frequency] on [Track B] — masking [Track A]"
```

Dynamic Imbalance Detection:

```
Compute RMS per track over 2-bar window
If RMS differential between any two active tracks > 6dB:
  → Trigger suggestion: "[Track] is [N]dB louder than [Track] — consider reducing"
```

Arrangement Conflict Detection:

```
Count tracks with peak energy above -6dBFS simultaneously
If count ≥ 3:
  → Trigger suggestion: "Multiple tracks peaking simultaneously — risk of masking"
```

Suggestion Format Specification

Every suggestion must include:

Field	Example	Requirement
Action	"Lower vocal by 2.1dB"	Specific, numeric where possible
Reason	"masking 800Hz–2kHz on synth lead"	Frequency range cited
Confidence	94%	Displayed as colored badge
Track reference	Violet chip with track name	Clickable — jumps to track
Accept button	Applies immediately	Green, primary action
Reject button	Dismisses, logs	Zinc, secondary
Preview button	Auditions before commit	Cyan, tertiary

Suggestion Panel UI

- Position: non-blocking — bottom center, above keyboard shortcut bar
- Width: 420px
- Height: auto based on suggestion count
- Maximum simultaneous suggestions displayed: 3 (oldest dismissed if 4th arrives)
- Dismiss on click-outside: yes
- Auto-dismiss: no — suggestions persist until explicitly acted on

---

8.4 LLPTE Core — Real-Time AI Engine

Package Architecture (from monorepo)

```
packages/
├── llpte-core/           # Pipeline orchestration, lifecycle, node wiring
├── llpte-adapters/       # inputRouter — multi-track routing, format normalization
├── llpte-signal/         # spectralAnalyzer — FFT, RMS, LUFS, true peak
├── llpte-ai/             # aiMixEngine — inference, confidence scoring, decisions
├── llpte-transition-graph/  # transitionGraph — curve precomputation, execution
```

```
└── llpte-execution/     # outputBus — final signal delivery, monitoring
```

Named Pipeline Nodes

```
// @llpte/llpte-core — pipeline definition
const pipeline: LLPTEPipeline = {
  nodes: {
    inputRouter:       new InputRouterNode(),    // @llpte/llpte-adapters
    spectralAnalyzer: new SpectralAnalyzerNode(), // @llpte/llpte-signal
    aiMixEngine:       new AIMixEngineNode(),    // @llpte/llpte-ai
    transitionGraph:   new TransitionGraphNode(), // @llpte/llpte-transition-graph
    outputBus:         new OutputBusNode(),       // @llpte/llpte-execution
  },
  edges: [
    { from: 'inputRouter',      to: 'spectralAnalyzer' },
    { from: 'spectralAnalyzer', to: 'aiMixEngine'      },
    { from: 'aiMixEngine',      to: 'transitionGraph'  },
    { from: 'transitionGraph',  to: 'outputBus'        },
  ],
};
```

Performance Contract (SLA)

Metric	Hard Limit	Current Measurement	Breach Action
Inference latency (p50)	≤15ms	10ms	Alert + throttle
Inference latency (p99)	≤25ms	TBD	Alert
Node tick time	≤1ms	0.8ms	Alert
Active edges	≤2000 (MVP)	847	Warning at 1500
Memory per node	≤50MB	TBD	Alert
Pipeline uptime	99.9% per session	—	Auto-restart

Demo Visibility Requirements (Non-Negotiable)

The following behaviors MUST be visible during any investor demo:

1. `inference 10ms` badge updating on LLPTE Core node every 100ms
2. Animated connector line dashes moving between all 5 nodes
3. Transition Graph tooltip: `847 active edges | last tick 0.8ms` on hover
4. Signal oscilloscope strip animated below graph (≥30fps)
5. LLPTE Core arc spinner rotating continuously during playback
6. Sparkline inside each node showing live output history

8.5 Time Savings Tracking

Data Model

```
interface SessionMetrics {
  sessionId: string;                    // UUID
  userId: string;                       // FK to users table
  startedAt: Date;
  endedAt: Date;
  totalDurationMs: number;
  aiActionsCount: number;               // Total AI decisions
  aiActionsAccepted: number;            // User confirmed
  aiActionsRejected: number;            // User dismissed
  aiActionsAutoApplied: number;         // Confidence ≥ 0.65
  manualAdjustments: number;            // Non-AI mixer interactions
  clippingEventsPrevented: number;      // VU clip events below threshold
  transitionsGenerated: number;
  transitionsAccepted: number;
  estimatedTimeSavedMs: number;         // Computed: AI actions × baseline manual t
  acceptanceRate: number;               // aiActionsAccepted / aiActionsCount
  baselineManualSessionMs: number;      // Rolling average from user history
  timeSavingPercent: number;            // (baseline - actual) / baseline
}
```

Time Savings Calculation Methodology

The `estimatedTimeSavedMs` field is computed using action-type benchmarks established from manual-only session baselines:

Action Type	Manual Baseline	AI Execution	Time Saved
Gain adjustment (one			44.5

track)	45 seconds	0.5 seconds	seconds
EQ sweep (one track)	90 seconds	1 second	89 seconds
Crossfade transition setup	180 seconds	5 seconds	175 seconds
Filter sweep transition	240 seconds	5 seconds	235 seconds
Frequency conflict detection	300 seconds (if found at all)	0.1 seconds	299 seconds

Baseline manual times are established from:

1. Initial values: benchmarked from 50-session manual-only alpha test
2. Per-user calibration: rolling average of manual session times before AI adoption

Display Format

Session Summary Panel (full-screen overlay at session end):

```
┌──────────────────────────────────────────────┐
│  Session Complete                            │
│                                              │
│  42% faster than your average               │
│  ──────────────────────────────────────     │
│  18 minutes saved this session              │
│                                              │
│  AI Actions       34                        │
│  Accepted         24  (71%)                │
│  Auto-Applied      8                        │
│  Manual Actions   12                        │
│                                              │
│  Clips Prevented   9                        │
│  Transitions       4                        │
│                                              │
│  [Export PNG]  [Export JSON]  [Close]       │
└──────────────────────────────────────────────┘
```

Inline Session Chip (top nav, during session):

- `AI: 34 actions` — zinc-800 background, violet text
- Updates every 30 seconds
- Clicking opens mini session summary popover

9. USER FLOWS — COMPLETE

Primary Flow: DJ Set Preparation

```
ENTRY POINT: User opens R3, lands on arrangement view

Step 1 — Track Load
├── Drag tracks from sidebar to arrangement
├── Each track auto-assigns a color based on spectral content
│   (bass-heavy = amber, high-freq = cyan, vocal = rose)
├── Waveforms render progressively as audio is analyzed
└── LLPTE pipeline initializes — node graph appears in Zone 4B

Step 2 — First Playback
├── Space bar → playhead begins from bar 1
├── LLPTE activates — node graph connectors animate
├── Spectral Analyzer begins FFT per frame
└── AI Mix Engine first decision within 2 bars

Step 3 — Auto-Leveling Fires
├── Toast: "AI Mix updated — confidence 94%"
├── Ghost knobs appear on affected channel strips
├── AI MIX track overlay renders prediction curve
└── User has 3 options:
    ├── Accept all → knobs animate to position
    ├── Accept per-track → ghost knob click to confirm
    └── Ignore → ghost knobs fade after 8 seconds

Step 4 — Frequency Conflict Detection
├── Suggestion panel appears: "Cut low-end on SYNTH LEAD below 80Hz"
├── Timeline marker appears at bar where conflict begins
├── User clicks Accept → EQ adjustment applied
└── Mix Suggestion logged, acceptance rate updated

Step 5 — Transition Zone Approach
├── Spectral Analyzer detects phrase boundary at bar 16
├── Transition Graph begins precomputation at bar 14
├── At bar 15: violet region overlay appears on timeline
├── Transition type selector popover appears
├── User selects: Key-Match Crossfade
├── Preview button auditions the transition
├── User accepts → transition committed to timeline
└── "Send to AI" successfully completed

Step 6 — Session End
├── Space bar → playback stops
```

```
├── Session Summary panel appears automatically
├── Stats: "42% faster — 18 minutes saved — 34 AI actions"
├── User clicks Export PNG → share-ready image generated
└── Metrics stored to PostgreSQL via Drizzle (sessionMetrics table)
```

Secondary Flow: Content Creator Quick Mix

```
ENTRY POINT: User drags 2 files — voice.wav + background_music.wav

Step 1 — Auto-Detection
├── LLPTE detects voice track (spectral profile: 80Hz–8kHz, transient speech patter
├── LLPTE detects music track (full spectrum, rhythmic pattern)
└── Suggestion fires immediately: "Duck music when voice is present"

Step 2 — AI Auto-Leveling
├── Voice track: normalized to -14 LUFS (broadcast standard)
├── Music track: set to -20 LUFS below voice (-6dB relative)
├── Confidence: 91%
└── Ghost knobs show proposed levels — user accepts

Step 3 — Frequency Masking Fix
├── Suggestion: "Cut music track at 800Hz–2kHz — competing with voice"
├── User accepts → EQ applied
└── Mix sounds instantly more professional

Step 4 — Export
├── Bounce to Audio → stereo WAV at 44.1kHz/24bit
└── Session summary: "3 AI actions — mix ready in 4 minutes"
```

Error Flow: LLPTE Inference Timeout

```
Step 1 — Inference latency exceeds 25ms (p99 breach)
├── LLPTE Core node: badge changes from green to amber
├── Badge text: "inference 26ms ⚠"

Step 2 — Auto-throttle
├── AI inference scope reduced to 4 tracks (from 8)
├── Toast: "AI scope reduced — high processing load"

Step 3 — Recovery
├── If latency returns to ≤15ms within 10 seconds:
│   └── Full scope restored, toast: "AI fully restored"
└── If latency sustained above 25ms:
    └── AI disabled for session, toast: "AI paused — reduce track count"
```

---

10. NON-GOALS (FIRM SCOPE BOUNDARY)

The following are explicitly excluded from MVP and any subsequent sprint until post-Series A:

Non-Goal	Why Excluded	When Revisited
VST/AU/AAX plugin support	Requires native binary bridge — months of work	Post-Series A
Advanced MIDI composition	Piano roll, CC automation, note editing — separate product surface	Roadmap Phase 3
Real-time collaboration	WebSocket multi-user is architecturally possible but requires conflict resolution logic	Roadmap Phase 2
Mobile platform	Audio engine requires desktop API access (AudioWorklet, SharedArrayBuffer)	Post $5M ARR
Sound design / synthesis	Granular, wavetable, FM synthesis — different user and use case	Never (separate product)
Plugin ecosystem (marketplace)	Requires SDK, certification, developer relations	Post-Series A
Hardware controller mapping	MIDI CC mapping requires device enumeration layer	Roadmap Phase 2
AI model training UI	Users should not need to train models — abstracted by LLPTE	Post-Series B
Offline mode	Cloud inference preferred for model updates — offline is a degraded fallback	Roadmap Phase 2

---

11. TECHNICAL REQUIREMENTS & ARCHITECTURE

Performance Targets (Full Specification)

Layer	Metric	Requirement	Measurement
Audio engine	Round-trip latency	≤10ms	AudioContext.currentTime delta
LLPTE inference	p50 latency	≤15ms	Pipeline tick timer
LLPTE inference	p99 latency	≤25ms	1000-iteration benchmark
Node tick	Per-node	≤1ms	Individual node timer
UI frame rate	During active playback	≥60fps	requestAnimationFrame timing
Waveform render	Initial load (8 tracks)	≤2 seconds	Time-to-paint measurement
WebSocket round-trip	State sync	≤50ms	Ping/pong timestamp
Database query	Session write	≤100ms	Drizzle query timing
File load	10MB audio file	≤1 second	ArrayBuffer load timing
Memory ceiling	8 tracks × 5 min	≤2GB RAM	process.memoryUsage()

Audio Engine Architecture

```
Web Audio API Context
├── AudioWorklet (WASM module — real-time processing)
│   ├── SharedArrayBuffer (lock-free ring buffer — no GC pauses)
│   ├── SIMD-accelerated DSP (Float32 SIMD via Wasm SIMD)
│   └── ≤3ms processing target (per PERF PRD)
├── AnalyserNode (FFT for LLPTE input)
├── GainNode × 8 (per track — controlled via AudioParam)
├── BiquadFilterNode × 3 per track (3-band EQ)
└── DynamicsCompressorNode (master bus)
```

WebGPU Rendering Architecture

```
WebGPU Device
├── OffscreenCanvas compute shaders (waveform rendering)
│   └── Target: ≤4ms per frame (per PERF PRD)
├── Typed array pool allocators (zero GC during playback)
```

```
└── React Three Fiber (3D visual engine — optional during MVP)
```

Network & Protocol

```
Client ←→ Server
├── tRPC (type-safe HTTP — all query/mutation procedures)
├── WebSocket (real-time: transport state, AI suggestions, session sync)
│   └── ws@8.20.0 — confirmed in node_modules
├── MessagePack binary protocol (LLPTE signal data — ≤50ms round-trip)
└── REST fallback for file upload (multer v2.1.1)
```

Database Architecture (Drizzle + PostgreSQL)

```
PostgreSQL
├── users (id, email, hashedPassword, createdAt, tier)
├── sessions (id, userId, startedAt, endedAt, metrics JSON)
├── sessionMetrics (id, sessionId, aiActionsCount, timeSavedMs, ...)
├── subscriptions (id, userId, tier, stripeCustomerId, status)
├── projects (id, userId, name, arrangementJSON, createdAt)
├── samples (id, userId, filename, path, duration, sampleRate, bitDepth)
└── aiDecisionLog (id, sessionId, actionType, confidence, accepted, timestamp)
```

Materialized views (per PERF PRD):

- `mv_user_session_averages` — baseline manual session times for Time Savings calculation
- `mv_ai_acceptance_rates` — rolling acceptance rate per user for confidence calibration

---

12. DATA ARCHITECTURE & SCHEMA

Core Type System (from `shared/` directory)

```
// shared/audio.types.ts
interface TrackConfig {
  id: string;
  name: string;
  color: TrackColor; // 'kick-red' | 'bass-amber' | 'synth-violet' | ...
  gainDb: number;
  muted: boolean;
```

```
  soloed: boolean;
  armedForRecord: boolean;
  clips: AudioClip[];
  automationLanes: AutomationLane[];
}

// shared/mixer.types.ts
interface ChannelStrip {
  trackId: string;
  gain: number;       // dB — pre-fader
  eq: {
    hi: number;       // ±15dB
    mid: number;      // ±15dB
    lo: number;       // ±15dB
  };
  faderDb: number;    // -∞ to +6dB
  vuLevel: number;    // dBFS — live
  peakHold: number;   // dBFS — held 2s
  aiGhostGain?: number;   // LLPTE suggestion
  aiGhostEq?: EQSuggestion; // LLPTE suggestion
  aiConfidence?: number;   // [0, 1]
}

// shared/dj.types.ts
interface CrossfaderState {
  position: number;  // [-1, 1] — -1 = hard left, 1 = hard right
  curve: 'linear' | 'exponential' | 'hamster';
  aiSuggestedPosition?: number;
}

// shared/effects.types.ts
interface EffectChain {
  trackId: string;
  effects: AudioEffect[];
}

interface AudioEffect {
  id: string;
  type: 'reverb' | 'delay' | 'compressor' | 'eq' | 'gate' | 'pitch';
  enabled: boolean;
  params: Record<string, number>;
}

// shared/waveform.types.ts
interface WaveformData {
  trackId: string;
  peaks: Float32Array;   // Peak amplitude per pixel
  rms: Float32Array;     // RMS per pixel
```

```
  sampleRate: number;
  duration: number;
}
```

LLPTE Decision Log Schema

```
// aiDecisionLog table
interface AIDecisionLog {
  id: string;           // UUID
  sessionId: string;    // FK
  timestamp: Date;
  nodeId: 'aiMixEngine' | 'transitionGraph';
  actionType: 'gain_adjust' | 'eq_suggest' | 'transition_generate' | 'conflict_flag
  trackId?: string;     // Affected track
  inputConfidence: number;   // Raw model confidence
  displayedConfidence: number; // After calibration
  decision: Record<string, unknown>; // JSONB — action-specific payload
  outcome: 'auto_applied' | 'accepted' | 'rejected' | 'ignored' | 'discarded';
  latencyMs: number;    // Inference time for this decision
}
```

---

13. API CONTRACT SPECIFICATION

tRPC Router Overview

```
// server/procedures.ts (primary)
const appRouter = router({
  // Auth
  auth: authRouter,
  // Projects
  projects: projectsRouter,    // CRUD for arrangements
  // Samples
  samples: samplesRouter,      // Browse, upload, preview
  // Sessions
  sessions: sessionsRouter,    // Start, end, metrics write
  // AI
  ai: aiRouter,                // LLPTE pipeline control
  // Subscriptions
  subscriptions: subsRouter,   // Tier management, Stripe
});
```

Key Procedure Contracts

ai.startPipeline

```
input: { sessionId: string; trackIds: string[] }
output: { pipelineId: string; nodeStatus: NodeStatus[] }
```

ai.getInferenceMetrics

```
input: { pipelineId: string }
output: {
  inferenceLatencyMs: number;
  activeEdges: number;
  lastTickMs: number;
  nodeMetrics: Record<string, NodeMetric>;
}
```

ai.submitSuggestionOutcome

```
input: {
  decisionId: string;
  outcome: 'accepted' | 'rejected' | 'ignored';
  manualValueIfOverride?: number;
}
output: { logged: boolean; acceptanceRateUpdated: number }
```

sessions.endSession

```
input: { sessionId: string }
output: {
  metrics: SessionMetrics;
  exportPngUrl?: string;  // Pre-signed S3/storage URL
}
```

WebSocket Event Contracts

```
// Client → Server
type ClientEvent =
  | { type: 'transport:play'; bpm: number }
  | { type: 'transport:stop' }
  | { type: 'ai:sendToAI'; clipId: string; transitionType?: TransitionType }
  | { type: 'suggestion:respond'; decisionId: string; outcome: Outcome };
```

```
// Server → Client
type ServerEvent =
  | { type: 'ai:suggestion'; suggestion: MixSuggestion }
  | { type: 'ai:levelingApplied'; trackId: string; gainDelta: number; confidence: n
  | { type: 'ai:transitionReady'; clipId: string; curveData: Float32Array }
  | { type: 'llpte:metricsUpdate'; metrics: LLPTEMetrics }
  | { type: 'session:timeSavedUpdate'; savedMs: number };
```

---

14. ANALYTICS & TELEMETRY

Event Schema (All Events)

Every analytics event must include:

```
interface BaseEvent {
  eventId: string;          // UUID
  userId: string;           // Hashed — never plaintext
  sessionId: string;
  timestamp: Date;
  appVersion: string;       // e.g., "4.2.1"
  platform: 'mac' | 'windows';
  tier: 'free' | 'pro' | 'studio';
}
```

Critical Events to Track

Event	Trigger	Key Properties
`session_started`	Playback begins	trackCount, bpm
`ai_suggestion_shown`	Suggestion appears in panel	actionType, confidence, trackId
`ai_suggestion_accepted`	User clicks Accept	decisionId, confidence, latencyMs
`ai_suggestion_rejected`	User clicks Reject	decisionId, confidence
`ai_auto_applied`	Confidence ≥ 0.65	decisionId, confidence, gainDelta
`transition_generated`	LLPTE generates	transitionType, confidence

	transition	
`transition_accepted`	User accepts transition	transitionType, edited: boolean
`send_to_ai_triggered`	Right-click → Send to AI	clipId, clipDurationMs
`time_savings_viewed`	Session summary opened	savedMs, savedPercent
`time_savings_exported`	Export PNG/JSON clicked	format
`upgrade_prompt_shown`	Free tier limit reached	limitType
`upgrade_completed`	Payment confirmed	fromTier, toTier
`llpte_latency_breach`	Inference >15ms	latencyMs, trackCount
`session_ended`	Playback stops	durationMs, totalAiActions

KPI Dashboard Requirements (for investors)

The following metrics must be queryable from the analytics database by any team member:

- Daily/Weekly/Monthly Active Users
- AI feature adoption rate (% of sessions with ≥1 AI action)
- Suggestion acceptance rate (rolling 7-day, 30-day)
- Average time savings per session (in minutes and percentage)
- Conversion rate: free → pro (by cohort)
- Day 1, 7, 30 retention by cohort
- LLPTE latency p50/p99 in production (24-hour rolling)
- Churn rate by tier

---

15. SECURITY & COMPLIANCE

Authentication Architecture

```
User Login
     │
     ├── Email/Password → bcrypt v6.0.0 (lazy-init, audit-confirmed)
     │                  → JWT (jsonwebtoken v9.0.3)
     │                  → Token stored in httpOnly cookie
     │
     └── Token Refresh → hydrateFromToken() race condition — FIXED
                       → Loading state guard — FIXED
                       → trpcAuth middleware mount — FIXED
```

Authorization Layers

Layer	Method	Enforcement Point
Route-level	`trpcAuth` middleware	`server/middleware/`
Procedure-level	tRPC context validation	`server/trpc.ts`
Data-level	userId FK on every query	Drizzle ORM — all reads filtered by userId
File access	Path traversal guard	Upload route — allowlist-validated
Subscription tier	Tier check in procedure	`server/procedures.ts` — all AI procedures

Known Security Fixes (Audit-Confirmed)

Issue	Status	Fix Location
Missing `trpcAuth` middleware mount	Fixed	`server/middleware/`
`hydrateFromToken()` race condition	Fixed	Auth initialization sequence
Missing loading state	Fixed	Auth context provider
Path traversal on uploads	Fixed	Upload route validator
`req.user` type augmentation	Fixed	`server/types/`
bcrypt lazy initialization	Fixed	Auth service startup
Admin seeder idempotency	Fixed	`server/scripts/`

Data Privacy

- User audio files: stored in `uploads/` — server-local for MVP, S3 post-MVP
- AI decision logs: stored with hashed userId — no plaintext PII in analytics events
- Session metrics: userId FK but no audio data — safe for analytics pipeline
- Subscription data: Stripe customer ID stored — card data never touches R3 servers
- JWT payload: userId + tier only — no email or PII in token

Rate Limiting

```
// express-rate-limit v7.5.1 — confirmed in node_modules
const aiRateLimit = rateLimit({
  windowMs: 60 * 1000,  // 1 minute
  max: 100,             // 100 AI requests per minute per user
  message: 'AI request limit reached — upgrade to Studio for higher limits',
});
```

---

16. ACCESSIBILITY

Requirements (WCAG 2.1 AA — Minimum)

Requirement	Implementation
Color contrast	All text on dark backgrounds: ≥4.5:1 ratio
Focus indicators	Visible focus ring on all interactive elements — `ring-2 ring-cyan-400`
Keyboard navigation	All controls reachable via Tab/Arrow keys
Screen reader labels	All icon buttons have `aria-label`
Motion sensitivity	All animations wrapped in `prefers-reduced-motion` media query
Audio control	All AI-applied audio changes are reversible via Ctrl+Z

AI-Specific Accessibility

- Every AI suggestion panel has an accessible dismiss button (keyboard: `Escape`)
- Ghost knob states announced via `aria-live="polite"` when they appear
- Confidence scores included in `aria-label` : e.g., `aria-label="Reverb suggestion — 94% confidence — Accept"`
- LLPTE node graph: static alternative text available via keyboard toggle for users who cannot perceive the animation
- VU meters: numerical dB values available via `title` tooltip for screen reader users

---

17. ERROR STATES & EDGE CASES

Audio Engine Errors

Error	Cause	UI Treatment	Recovery
AudioContext suspended	Browser requires gesture to start audio	Full-screen overlay: "Click to enable audio"	User clicks → AudioContext.resume()
Sample load failure	Network error or corrupt file	Toast: "Failed to load [filename] — try re-uploading"	Retry button in toast
Buffer underrun	CPU overload	Toast: "Audio dropout detected — reducing AI scope"	Auto-throttle AI pipeline
SharedArrayBuffer unavailable	Missing COOP/COEP headers	Fallback to non-WASM audio engine	Silent degradation, lower performance
WebGPU unavailable	Unsupported browser	Fallback to Canvas2D rendering	Silent degradation, 2D waveforms only

LLPTE Pipeline Errors

Error	Cause	UI Treatment	Recovery

Inference timeout (>25ms)	High load	Node badge: amber `inference 26ms ⚠`	Auto-throttle — reduce to 4 tracks
Node initialization failure	Package load error	Toast: "AI pipeline failed to start — restart required"	Reload button
Confidence below threshold for all suggestions	Unusual audio content	No suggestions shown — no error	Expected behavior, log for model improvement
Transition precompute failure	Spectral data incomplete	Fallback to manual transition mode	Toast: "AI transition unavailable — manual mode active"

Database Errors

Error	Cause	UI Treatment	Recovery
Session write failure	DB connection lost	Local session data buffered, retry on reconnect	Retry with exponential backoff
Subscription tier query failure	DB unavailable	Fallback to previously cached tier	Show stale badge, retry in background
Project save failure	Network error	Toast: "Save failed — retrying..." with unsaved indicator	Auto-retry 3 times, then prompt manual save

Subscription & Auth Errors

Error	Cause	UI Treatment	Recovery
JWT expiry during session	Token expired	Silent refresh attempt	If refresh fails: soft sign-out with save prompt
Stripe webhook failure	Payment processor timeout	Tier unchanged until resolved	Admin alert, user email notification
Free tier limit	User exceeded	Non-blocking modal:	Dismissed without data

reached	track/AI limit	upgrade prompt	loss
Invalid subscription tier	DB/Stripe desync	Default to free tier	Admin reconciliation queue

---

18. ONBOARDING FLOW

First Session — New User

```
Step 1: Account Creation
├── Email + password form (or OAuth — future)
├── "What will you use R3 for?" — single-select:
│   ├── DJ mixing / set prep
│   ├── Content creator (YouTube, Podcast)
│   └── Music production / beats
└── Tier selection: Start Free or Try Pro (14-day trial)

Step 2: Empty State — First Time in Arrangement View
├── Large centered prompt: "Drop your first track to begin"
├── Sample pack offered: "Try with our sample set →"
│   └── Pre-loaded 8-track demo session — instantly demonstrates AI
└── Tutorial tooltip sequence (dismissable, keyboard: Escape)

Step 3: First Track Loaded
├── Tooltip 1: "This is your arrangement view — tracks live here"
├── Waveform renders automatically
└── Tooltip 2: "Press Space to play — R3's AI starts listening"

Step 4: First Playback
├── AI Auto-Leveling fires within 2 bars
├── Toast: "AI Mix updated — confidence 94%"
├── Tooltip 3: "That's your first AI suggestion — click to accept"
└── Ghost knob appears on mixer

Step 5: First Accept
├── Knob animates to new position
├── Tooltip 4: "AI adjusted your gain — mix sounds cleaner"
└── Tooltip 5: "Right-click any clip to Send to AI for transitions"

Step 6: First Session End
├── Session Summary panel appears
├── Stats shown even if minimal (first session)
```

```
└── CTA: "Upgrade to Pro to unlock all AI features →"
```

Onboarding Completion Criteria

A user is considered “activated” (past the onboarding funnel) when:

- They have completed ≥1 full session (played + stopped)
- They have accepted ≥1 AI suggestion
- They have seen the Time Savings Summary panel

Target: 70% of new users reach activation within first session.

---

19. EFFECTS RACK — FULL SPECIFICATION

Reverb Module

Algorithm: Algorithmic reverb (Schroeder-Moorer variant) implemented in AudioWorklet

Parameter	Range	Default	LLPTE Control
Room Size	0–100%	50%	Read-only (AI reads, suggests, does not auto-set)
Decay	0.1–10s	3.2s	Ghost value shown when AI suggests
Wet/Dry	0–100%	28%	Same
Pre-delay	0–100ms	20ms	Manual only
Damping	0–100%	40%	Manual only

Visualization: Frequency response curve — translucent exponential decay in `rgba(59, 130, 246, 0.3)` on zinc-950 canvas. Updates when parameters change.

Delay Module

Algorithm: Stereo ping-pong delay with tempo sync

Parameter	Range	Default	Notes
Time	1/128 to 1/1	1/8	Tempo-synced divisions

Feedback	0–95%	45%	95% max — prevents infinite feedback
Spread	0–100%	80%	L/R separation
Filter	LP/HP cutoff	Flat	Pre-delay filter

Visualization: Two dots animating between L and R channels at the delay time interval. Tempo-locked via AudioContext.currentTime.

Compressor Module

Algorithm: Web Audio API DynamicsCompressorNode with extended parameter control

Parameter	Range	Default	Notes
Threshold	−60 to 0dBFS	−18dB	LLPTE reads for masking detection
Ratio	1:1 to ∞:1	4:1	Inf = hard limit
Attack	0.1–100ms	5ms	
Release	10–1000ms	80ms	
Knee	0–40dB	3dB	Soft knee
Makeup Gain	0–24dB	Auto	Auto-gain compensation

Visualization: Gain reduction meter — shows `-3.2dB GR` in amber `#F59E0B` when active. Updates at 60fps during playback.

Parametric EQ Module

Bands: Low shelf + 3× parametric bell + High shelf = 5 bands total

Band	Type	Default Frequency	Default Gain
Band 1	Low Shelf	80Hz	+3dB (kick body boost)
Band 2	Bell	300Hz	0dB
Band 3	Bell	3kHz	−2dB (harshness cut)
Band 4	Bell	8kHz	−1.5dB (sibilance reduction)
Band 5	High Shelf	12kHz	0dB

Visualization: 80×40px mini frequency curve on zinc-950, rendered in cyan `#00F5FF`. Interactive: click+drag bands to adjust. Full-size view in expand mode.

Gate Module

Parameter	Range	Default
Threshold	−80 to 0dBFS	−40dB
Attack	0.1–100ms	1ms
Hold	0–500ms	50ms
Release	10–1000ms	200ms
Range	0 to −90dB	−60dB

Visualization: Mini oscilloscope — gated portions rendered as flat line at range value, active signal rendered in full color. Demonstrates gate behavior clearly.

Pitch Shifter Module (Bypassed by Default)

Parameter	Range	Default
Semitones	−24 to +24	0
Fine	−100 to +100 cents	0
Formant Preservation	On/Off	On

Visualization: Piano-key strip (2 octaves) showing current pitch offset. Keys above root highlighted in cyan, below in amber.

Demo Note: This module ships bypassed. Never activate it during an investor demo without deliberate intent — pitch artifacts are the most jarring possible audio failure in a live demo.

---

20. PERFORMANCE BENCHMARKING METHODOLOGY

LLPTE Latency Benchmark

```
// Benchmark script — run before any demo or release
async function benchmarkLLPTE(iterations: number = 1000) {
  const latencies: number[] = [];
  for (let i = 0; i < iterations; i++) {
    const start = performance.now();
    await pipeline.tick(mockAudioFrame);
    latencies.push(performance.now() - start);
  }
  return {
    p50: percentile(latencies, 50),
    p90: percentile(latencies, 90),
    p99: percentile(latencies, 99),
    max: Math.max(...latencies),
    mean: mean(latencies),
  };
}
// Pass criteria: p50 ≤15ms, p99 ≤25ms, max ≤50ms
```

Audio Dropout Detection

```
// Integrated in AudioWorklet — fires on every process() call
const DROPOUT_THRESHOLD_MS = 10;
process(inputs, outputs) {
  const start = performance.now();
  // ... processing ...
  const elapsed = performance.now() - start;
  if (elapsed > DROPOUT_THRESHOLD_MS) {
    this.port.postMessage({ type: 'dropout', elapsedMs: elapsed });
  }
}
```

UI Frame Rate Monitor

```
// Runs during QA — confirms 60fps during playback
let lastFrame = performance.now();
function frameMonitor() {
  const now = performance.now();
  const fps = 1000 / (now - lastFrame);
  lastFrame = now;
  if (fps < 55) console.warn(`Frame rate drop: ${fps.toFixed(1)}fps`);
  requestAnimationFrame(frameMonitor);
```

```
}
```

Pre-Demo QA Checklist

Run this 30 minutes before any investor meeting:

- LLPTE benchmark: p50 ≤15ms confirmed
- Node graph: all 5 nodes rendering with animated connectors
- `inference 10ms` badge: updating every 100ms
- Transition Graph tooltip: shows active edge count
- Ghost knobs: appearing within 2 bars of playback start
- Toast notifications: rendering and auto-dismissing
- Time Savings panel: populating with real data
- Export PNG: generating correctly
- "Send to AI": context menu functioning
- All 6 effects modules: loading without errors
- VU meters: active on all channel strips
- Crossfader: responding to mouse drag
- Session summary: appearing on playback stop
- Pro badge: visible in top nav
- CPU meter: below 60% at 8 tracks
- Audio: no dropouts in 5-minute sustained playback test
- Memory: below 2GB at 8 tracks

---

21. RISKS & MITIGATIONS (EXPANDED)

Risk Registry

ID	Risk	Probability	Impact	Mitigation	Owner

R1	AI feels gimmicky	Medium	Critical	Ship only features with measurable, audible impact. Auto-Leveling first — produces instant verifiable result.	Product
R2	LLPTE latency degrades under load	Medium	High	Confidence gating + auto-throttle. 8-track limit. Per-node tick monitoring live in UI.	Engineering
R3	User distrust of AI decisions	Medium	High	Ghost knobs + confidence scores + easy override. Never auto-apply below 0.65.	Product
R4	Demo crashes in front of investor	Low	Critical	Pre-demo QA checklist (Section 20). Demo environment hardcoded Pro tier, billing disabled.	Engineering
R5	Audio artifacts from AI adjustments	Low	High	All gain changes via AudioParam.setTargetAtTime() — click-free by design. QA: 48-hour listening test.	Engineering
R6	Competitive response from Ableton/NI	Low	Medium	LLPTE is architectural, not cosmetic — cannot be feature-flagged into a legacy DAW. 12-month moat minimum.	Strategy
R7	Inference cost exceeds revenue	Medium	Medium	Local inference for MVP — no cloud cost. Post-MVP: quantized models on-device.	Engineering
R8	Subscription churn before habit forms	High	High	Day 1 onboarding targets activation in first session. Habit = 3 sessions/week.	Growth
R9	Creator economy market shifts	Low	Low	Wedge strategy — DJs are the primary target. Creator segment is upside, not dependency.	Strategy
R10	WebGPU browser compatibility	Medium	Low	Canvas2D fallback implemented. MVP does not require WebGPU for core functionality.	Engineering

22. MVP DEFINITION — INVESTOR CHECKLIST

Functional Requirements

Requirement	Status	Test
Load and play ≥8 simultaneous tracks	Required	8-track test session
Colored waveform rendering per track	Required	Visual QA
Cyan playhead — bar-accurate sync	Required	AudioContext timing audit
LLPTE node graph — animated, live inference timing	Required	Node badge updates every 100ms
AI Auto-Leveling — active within 2 bars	Required	Automated playback test
Ghost knobs visible on mixer strips	Required	Visual QA
Confidence score on AI MIX track	Required	`94%` minimum in demo
"Send to AI" context menu	Required	Right-click on any clip
≥1 transition type (crossfade)	Required	Execute and preview
Time Savings panel with real data	Required	Session end trigger
Toast notifications	Required	Toast fires on AI action
VU meters active on all 6 channel strips	Required	Visual QA during playback
Effects rack: Reverb, Delay, Compressor minimum	Required	Knob interaction
Keyboard shortcut bar visible	Required	Visual QA
Pro badge in top nav	Required	Subscription tier check

Demo-Specific Requirements

Requirement	Status	Failure Condition

`inference 10ms` badge updating live	Required	Static badge = demo failure
Transition Graph tooltip: edge count + tick time	Required	No tooltip = demo failure
Signal oscilloscope animated	Required	Static waveform = demo failure
Session summary exportable PNG	Required	Export fails = demo failure
No audio dropouts in 20-minute demo	Required	Any dropout = demo failure
No crashes	Required	Any crash = demo failure
AI suggestion fires in first 2 bars	Required	No suggestion = demo failure

---

23. POST-MVP ROADMAP

Phase 2 — Q3 2026 (6 months post-launch)

Theme: Depth + Community

Feature	Description	Rationale
Stem separation	AI-powered track separation (vocals, drums, bass, other) from mixed audio	Unlocks mashup workflow for DJs
Custom AI preferences	User can weight AI suggestions by genre/style	Reduces false positives for experienced users
Session history	Browse, replay, and fork past sessions	Retention — users can revisit and improve past work
Hardware controller mapping	MIDI CC to mixer/effects parameters	Bridges gap between R3 and existing DJ hardware
Collaboration (beta)	Real-time multi-user session sharing	Unlocks studio tier use cases

Phase 3 — Q1 2027 (12 months post-launch)

Theme: Platform + B2B

Feature	Description	Rationale
LLPTE API (public)	tRPC endpoints exposed for third-party integrations	B2B revenue — DAW plugins, hardware
Mobile companion	iOS/Android remote transport control and monitoring	Extend the studio to mobile without rebuilding the core
Advanced analytics	Per-genre AI calibration, personalized baseline	Model improves with user data flywheel
Label/studio tier	Multi-user workspace with admin controls	Enterprise contract opportunity
Plugin support (limited)	VST3 receive-only — no authoring	Opens R3 to existing producer workflows

Phase 4 — Q3 2027 (24 months post-launch)

Theme: AI Model + Market Expansion

Feature	Description	Rationale
R3 model fine-tuning	User can fine-tune LLPTE on their own mix history	True personalization — highest retention driver
Generative transitions	AI generates entirely new transition content, not just curves	Category-defining feature
Hardware partnerships	R3 software bundled with DJ controller manufacturers	Distribution channel at hardware scale
Marketplace	Community-shared AI presets, transition styles, effect chains	Network effects — platform moat

---

24. GO-TO-MARKET STRATEGY

Channel Strategy

Channel 1: Content-Led (Primary — 0 to $500K ARR)

The most cost-effective channel is demonstrating the product through the product. R3's visual identity — the LLPTE node graph, the ghost knobs, the Time Savings panel — is inherently content-native. It looks good on a screen recording.

Content Formats:

- "Before AI / After AI" side-by-side audio comparisons (60 seconds)
- LLPTE node graph explanations — "This is what's actually happening when R3 mixes your tracks"
- Time Savings reveal videos — "R3 just told me I mixed that set 42% faster"
- Persona-specific content — Marcus prep'ing a 2-hour DJ set in 45 minutes

Distribution:

- Twitter/X (audio production community)
- TikTok (creator community)
- YouTube Shorts (tutorial adjacent)
- Reddit: r/DJs, r/WeAreTheMusicMakers, r/edmproduction, r/podcasting

KPI: 100 signups per viral content piece at >100K views.

Channel 2: Community (Primary — 0 to $500K ARR)

DJ Community:

- DJTT (DJ Tech Tools) — editorial relationship + sponsored content
- DJ forums — authentic participation, not advertising
- Discord servers — DJ communities where Marcus lives

Creator Community:

- Creator economy newsletters (Matt Gielen, Paddy Galloway)
- Podcast communities (r/podcasting, Podcast Movement)

KPI: 500 alpha users from community channels in first 90 days.

Channel 3: Hardware Partnerships (Medium-term — $500K to $3M ARR)

DJ controllers are purchased by DJs. The moment of purchase is the highest-intent moment for DJ software adoption. Pioneer DJ, Rane, Denon DJ, and Numark collectively ship 3M+ units annually.

Approach: Bundle R3 Pro (3-month trial) with mid-tier DJ controller purchases via OEM agreement.

Projected impact: 10% of 500K relevant units shipped annually = 50K new users. At 10% conversion to paid: 5,000 new subscribers = $100K MRR.

Channel 4: Paid Acquisition (Late-stage — >$1M ARR)

Paid social and search only after organic channels establish creative benchmarks. Do not pay for acquisition before organic CAC is measured.

Target CAC: ≤$35 via content. ≤$60 via paid social.

---

25. FINANCIAL MODEL (HIGH-LEVEL)

Cost Structure

Category	Monthly Cost (Pre-Scale)	Notes
Infrastructure (Railway/Docker)	$200	Scales with users
PostgreSQL (managed)	$100	
AI inference (local — MVP)	$0	No cloud inference cost
Storage (S3 equivalent)	$50	Audio file uploads
Stripe fees	2.9% + $0.30 per transaction	
Error monitoring (Sentry)	$26	
Analytics	$50	
Total fixed/semi-fixed	**~$450/month**	

At 500 paying users ($20/month): Revenue $10K, Infrastructure ≈ $600, Gross Margin ≈ 94%.

At 5,000 paying users: Revenue $100K, Infrastructure ≈ $3,000, Gross Margin ≈ 97%.

Funding Ask (Context)

The current build state — LLPTE implemented, UI shipped, 42 Vitest cases passing, auth fixed, database schema defined — represents approximately $300–600K of technical value as a pure asset.

With a working demo and 50+ beta users: $800K–$2.5M pre-seed. With PRD success metrics achieved (≥65% AI suggestion acceptance, 50–100 paying beta users): $3–6M seed.

The gap between current state and seed metrics is approximately 4–6 months of focused execution.

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26. INVESTOR DEMO SCRIPT

Setup (30 minutes before)

1. Run pre-demo QA checklist (Section 20) — all 17 items confirmed
2. Load demo session: 8 pre-selected tracks, 128 BPM, key of A minor
3. Confirm Pro badge visible in top nav
4. Confirm LLPTE node graph visible and animated
5. Dim room lights — screen is the only light source for maximum visual impact
6. Have Time Savings export ready from a prior session to show the panel immediately if needed

Demo Script (12 minutes)

Minute 0–1: The Hook

> “I’m going to show you what happens when you give a DJ’s workflow an AI brain. Watch the bottom-right corner.” Press Space. LLPTE activates. Node graph animates. Investor’s eyes go to the node graph.

Minute 1–2: First AI Action

> “Within two bars, R3 has already analyzed all 8 tracks simultaneously.” Toast fires: “AI Mix updated — confidence 94%” “That’s not a notification. That’s the AI telling you it just made 6 gain adjustments. See the ghost knobs on the mixer?” Point to ghost knob positions on channel strips.

Minute 2–4: Accept and Override

> “I can accept all of these—” click Accept “—watch the knobs move to their new

> positions." Knobs animate. "Or I can override any of them. The AI learns from every override." Demonstrate one manual override.

Minute 4–6: Frequency Conflict

> "Now this is where it gets interesting." Suggestion panel fires: "Cut low-end on SYNTH LEAD below 80Hz — masking 808 BASS" "The AI just caught a frequency clash that a beginner would not hear for another 20 minutes of listening. A professional would catch it in 2. R3 caught it in 6 seconds." Click Accept.

Minute 6–8: Send to AI / Transition

> Right-click VOCAL CHOP → Send to AI "This is our 'Send to AI' transition workflow." Violet overlay appears. Transition type selector appears. Select Key-Match Crossfade. "The AI has already analyzed the harmonic relationship between these tracks using the Camelot wheel." Click Preview — transition auditions. "Sounds clean. Accept." Transition committed.

Minute 8–10: The Node Graph

> "Let me show you what just happened under the hood." Hover over Transition Graph node — tooltip appears: `847 active edges | last tick 0.8ms` "That is 847 active decision edges processing at 0.8 milliseconds per tick. Our inference latency is 10 milliseconds. The best professional AI inference pipelines hit 15 milliseconds. We're at 10." Pause.

Minute 10–12: Time Savings

> Press Space to stop. Session Summary panel appears. "42% faster than the manual baseline. 18 minutes saved. 34 AI actions. 71% acceptance rate." Click Export PNG. "That image just got generated. A DJ can post that. An investor can put it in a deck." Pause. "This is not a feature. This is a new category."

---

27. GLOSSARY

Term	Definition
LLPTE	Low-Latency Processing Transition Engine — R3's named real-time AI pipeline, implemented as 6 TypeScript monorepo packages
inputRouter	LLPTE node 1 — routes multi-track audio into the pipeline, normalizes format
spectralAnalyzer	LLPTE node 2 — performs FFT, RMS, LUFS, and true peak

	measurement per frame
aiMixEngine	LLPTE node 3 — inference engine, produces gain/EQ decisions with confidence scores
transitionGraph	LLPTE node 4 — precomputes and executes transition curves; 847 active edges in demo
outputBus	LLPTE node 5 — final signal delivery and monitoring
Ghost knob	UI element showing where the AI wants to move a mixer parameter — translucent indicator layered on top of the actual knob position
Confidence score	AI's self-assessed probability that a decision is correct — displayed as a percentage; gate at 0.65 for auto-apply
Confidence gating	Logic that determines whether a decision is auto-applied (≥0.65), surfaced as a suggestion (≥0.40), or discarded (<0.40)
Send to AI	Right-click context menu action that routes a selected audio clip into the LLPTE pipeline for transition generation
Camelot wheel	Harmonic compatibility scoring system for DJ mixing — R3 uses it for key-match crossfade confidence scoring
Time Savings panel	Session summary overlay showing quantified productivity gains — primary investor demo artifact
DXA	Twentieths of a point — measurement unit used in OOXML for Word document layout (not used in R3 UI, used in docx export)
Vitest	TypeScript-native test framework used for LLPTE test coverage — 42 cases confirmed across AI Auto-Leveling and Smart Transitions
LUFS	Loudness Units Full Scale — broadcast-standard loudness measurement (ITU-R BS.1770-4) used by AI Auto-Leveling
RMS	Root Mean Square — average signal energy over a time window, used for dynamic imbalance detection
True peak	Inter-sample peak measurement — catches digital clipping that sample-level peak metering misses
Phrase boundary	The musical point where one section ends and another begins — detected by spectralAnalyzer for transition triggering
Ghost knob	See above

tRPC	Type-safe Remote Procedure Call framework — all client-server communication in R3 uses tRPC
Drizzle ORM	TypeScript-native database query builder used for all PostgreSQL access in R3
AudioWorklet	Web API for real-time audio processing in a dedicated high-priority thread — R3's audio engine core
SharedArrayBuffer	Lock-free memory buffer shared between AudioWorklet and main thread — enables zero-copy audio data transfer
WebGPU	Web API for GPU-accelerated compute and rendering — used for waveform rendering via OffscreenCanvas compute shaders
Zinc-950	Tailwind CSS color token — the darkest background in R3's design system (`#09090b`)
JetBrains Mono	Monospace typeface used for all data readouts in R3 (BPM, dB, ms, Hz, confidence scores)
Inter	Sans-serif typeface used for all labels, navigation, and prose in R3

FINAL PRINCIPLE

This document is a contract, not a wish list.

Every section is grounded in what is built, what is partially built, or what has a clear engineering path to completion. Nothing here is speculative architecture — the LLPTE pipeline runs, the UI is shipped, the tests pass.

The goal remains what it was in version 1.0 of this PRD:

Prove that AI can reduce the time and skill required to create a clean, professional mix — in real time.

The LLPTE pipeline makes that proof visible. The Time Savings panel makes it quantifiable. The ghost knobs make it trustworthy. The 10ms inference badge makes it undeniable.

That is the leverage investors fund. That is the product R3 v4 is becoming.